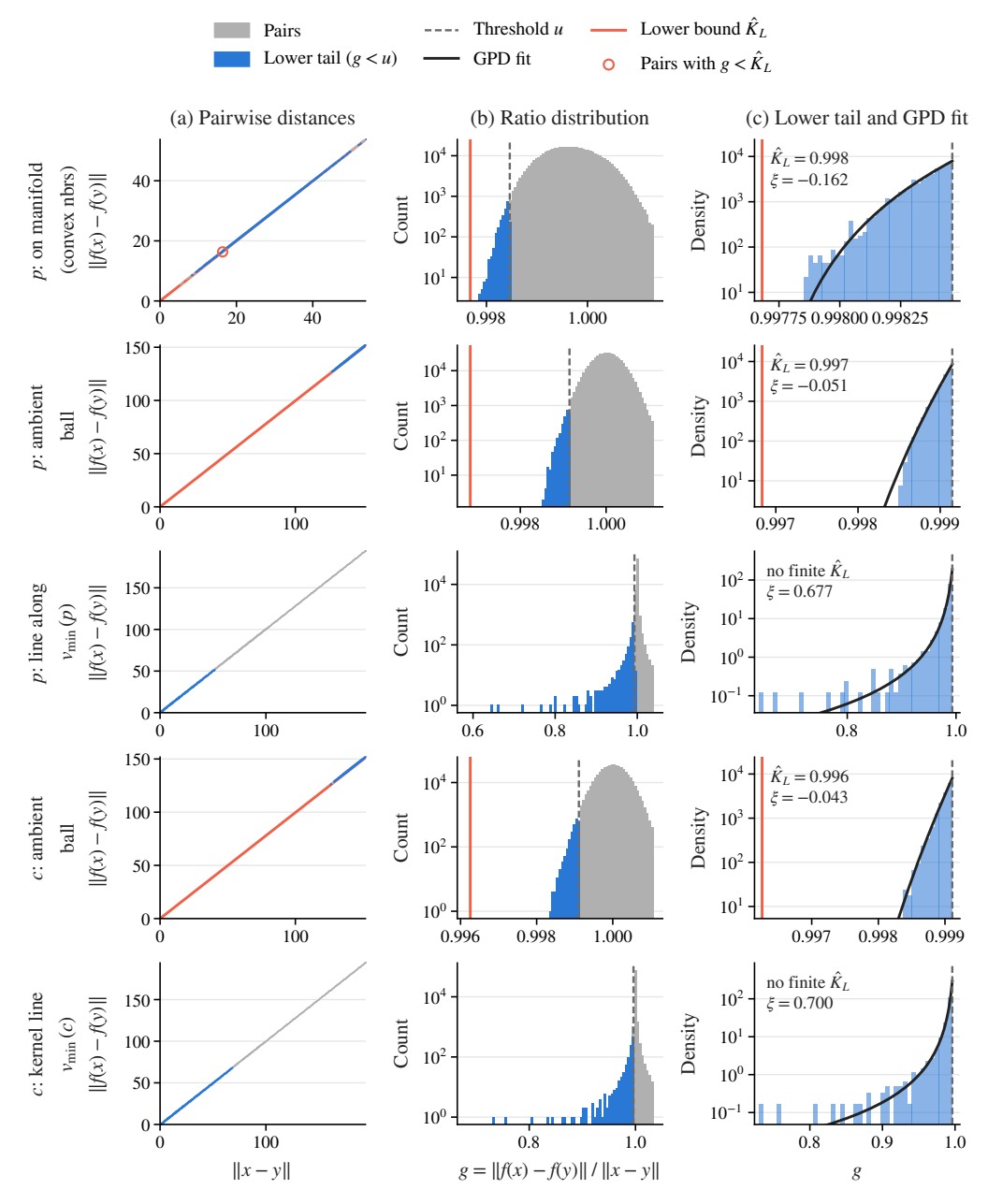

Pairs
Lower tail ($g < u$)
Threshold u
GPD fit
Lower bound $\hat{K}_L$
Pairs with $g < \hat{K}_L$
(a) Pairwise distances
(b) Ratio distribution
(c) Lower tail and GPD fit
p: on manifold (convex nbrs)
p: ambient ball
p: line along $v_{\min}(p)$
c: ambient ball
c: kernel line $v_{\min}(c)$
$\|f(x) - f(y)\|$
Count
Density
$\hat{K}_L = 0.998$
$\xi = -0.162$
$\hat{K}_L = 0.997$
$\xi = -0.051$
no finite $\hat{K}_L$
$\xi = 0.677$
$\hat{K}_L = 0.996$
$\xi = -0.043$
no finite $\hat{K}_L$
$\xi = 0.700$
$\|x - y\|$
$g = \|f(x) - f(y)\| / \|x - y\|$
g